

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	D.Raghava
Register Number	192525304

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Raghava.D(192525304) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.RAGHAVA

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premises computing.

A) Cloud Computing: Cloud computing is a technology that provides computing resources such as servers, storage, databases, networking, software, and processing power over the Internet whenever and wherever they are needed. Instead of purchasing and maintain expensive physical hardware in an organization, users can access these resources from cloud service providers and pay based on their usage.

*** Four Characteristics That Make Cloud Computing Different from Traditional On-Premises counting**

1. On-Demand Self-Service:

In cloud computing, **users can create or access resources such as virtual machines, storage, and applications** whenever they need them, without waiting for an IT administrator.

Example: A company can create a new server within minutes using the cloud, whereas traditional computing may require purchasing, installing, and configuring physical hardware.

2. Rapid Elasticity and Scalability:

Cloud resources can be **increased or decreased quickly** according to demand. This means the cloud can automatically handle sudden increases or decreases in workload.

Example: During an online sale, a shopping website can instantly add more servers to handle millions of users and reduce them after the sale.

3. Resource Pooling:

Cloud providers use a shared pool of computing resources such as servers, storage, and networks to serve multiple customers. Resources are dynamically assigned according to the user's requirements.

Example: Just like passengers share a bus instead of each person owning a bus, multiple customers can share cloud infrastructure while their data remains securely separated.

4.Measured Service / Pay-As-You-Go:

Cloud usage is continuously monitored and measured. Users generally pay only for the resources they consume, similar to an electricity bill.

Example: If a business uses 100 GB of storage and 10 hours of computing power, it pays according to its usage rather than purchasing an entire data center.

Creative Comparison: Traditional Computing vs Cloud Computing

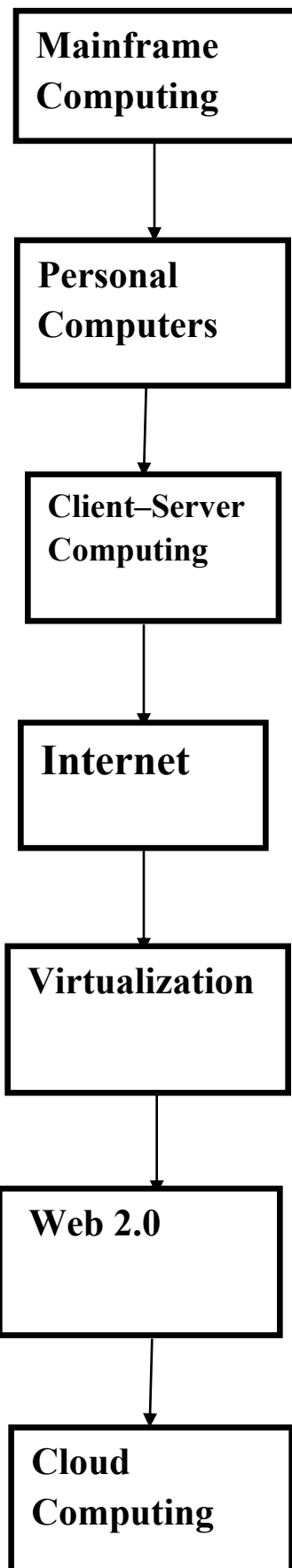
Traditional On-Premises Computing	Cloud Computing
Buy and maintain physical servers	Access virtual resources through the Internet
High initial investment	Pay according to usage
Scaling requires new hardware	Scale resources within minutes
Organization manages the infrastructure	Cloud provider manages the infrastructure
Resources may remain unused	Resources are dynamically allocated

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution and show how these changes enabled modern cloud platforms.

A) Evolution of Cloud Computing:

Cloud computing did not appear suddenly. It evolved gradually through major developments in hardware, networking, virtualization, the Internet, and software technologies. Each stage solved the limitations of the previous stage and finally led to today's modern cloud platforms.

Timeline:



1. Hardware Evolution: Mainframes and Time-Sharing (1960s–1970s):

In the early days, computers were extremely large, expensive, and available only to large organizations. Many users accessed a single powerful mainframe computer through terminals. The concept of time-sharing allowed multiple users to share the processing power of one computer.

2. Personal Computers and Distributed Computing (1980s):

The development of smaller and more affordable **personal computers (PCs)** allowed individuals and organizations to own their own computing systems.

However, each computer operated mostly independently, which resulted in:

- High hardware costs
- Difficult maintenance
- Limited resource sharing

Later, **distributed computing** connected multiple computers to work together.

3. Client–Server Architecture and Networking (1980s–1990s):

The **client–server model** divided computing tasks between:

- **Clients** – computers used by end users
- **Servers** – powerful computers that provided services and data

Organizations began using centralized servers for file storage, applications, and databases\

4. Internet and World Wide Web Evolution (1990s):

The rapid growth of the **Internet** and the **World Wide Web** made it possible to access information and applications from anywhere.

Initially, websites mainly provided static information. Later, dynamic web applications enabled users to interact with online services.

5. Virtualization Technology (Late 1990s–2000s):

Virtualization was one of the most important developments in cloud computing. A single physical server could be divided into multiple **virtual machines (VMs)**.

6. Web 2.0 and Internet Software Evolution (2000s):

Web applications became more interactive, dynamic, and service-oriented. Technologies such as **Service-Oriented Architecture (SOA)**, **APIs**, **web services**, and **distributed software** allowed applications to communicate over the Internet.

7. Modern Cloud Platforms:

With the combination of: **Powerful hardware + High-speed Internet + Virtualization + Distributed software + Automation** cloud platforms became possible.

Modern cloud providers offer services such as:

- **IaaS** – Virtual machines, storage, and networking
- **PaaS** – Platforms for application development
- **SaaS** – Ready-to-use software applications
- **Serverless Computing** – Running code without managing servers

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization in Cloud Computing:

Server virtualization is the process of dividing one physical server into multiple virtual machines (VMs) using a software layer called a hypervisor. Each VM can run its own operating system and applications independently.

Role in cloud computing:

- Improves utilization of physical servers
- Reduces hardware and operating costs
- Enables resource sharing
- Supports rapid scaling and flexibility
- Allows quick creation and management of virtual servers

Difference:

Virtualization	Cloud Computing
It is an enabling technology .	It is a service delivery model .
Creates virtual resources from physical hardware.	Provides computing resources over the Internet.
Mainly uses hypervisors and virtual machines.	Uses virtualization along with networking, storage, automation, and management.
Example: Creating multiple VMs on one physical server.	Example: Renting a virtual server from a cloud provider on a pay-as-you-use basis.

4)Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

SOAP: A strict, XML-based protocol that uses structured messages and is relatively heavyweight.

REST: A lightweight architectural style that uses HTTP methods and commonly exchanges JSON data.

For cloud applications, REST is generally preferred because it is simpler, faster, flexible, and easier to scale. SOAP is preferred when strict security and reliability standards are required.

SOAP vs REST Web Services

Feature	SOAP	REST
Communication style	Uses strict XML-based messaging and defined operations.	Uses HTTP methods such as GET, POST, PUT, and DELETE.
Data exchange	Mainly uses XML .	Commonly uses JSON , XML, or other formats.
Cloud suitability	More complex and heavyweight.	Lightweight, flexible, and highly suitable for cloud applications.

5) Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Sol)

IaaS, PaaS, SaaS and XaaS:

- **IaaS:** Provides virtual infrastructure. Example: Amazon EC2.
- **PaaS:** Provides a platform for application development. Example: Google App Engine.
- **SaaS:** Provides ready-to-use software online. Example: Gmail.
- **XaaS:** Provides any IT service through the cloud. Example: Dropbox (Storage as a Service).

IaaS, PaaS, SaaS and XaaS

Model	Meaning	Example
IaaS	Provides virtual infrastructure like servers and storage.	Amazon EC2
PaaS	Provides a platform to develop and deploy applications.	Google App Engine
SaaS	Provides ready-to-use software through the Internet.	Gmail
XaaS	Provides any IT resource as a service.	Dropbox (Storage as a Service)

